# Supervyse OPF Web UI Developer's Guide

***Revision 1.0***

***Aug 2024***

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# Revision History

| Revision Number | Date | Author | Reviewer | Summary |
|---|---|---|---|---|
| 1.0 | 2023-03-20 | Chijung Lin | Chijung Lin | Create this document provide a guide for web UI customization (included change menu item and change logo/resources/..). |
| 1.1 | 2023-06-30 | Chijung Lin | Chijung Lin | Review content for change menu style. |
| 1.2 | 2023-07-06 | Jamey Hou | Chijung Lin | Add customization menu style. |
| 1.3 | 2024-03-25 | Chijung Lin | Chijung Lin | Update the sample case for changing logo. |
| 1.4 | 2024-11-11 | Jamey Hou | Ethan Tesng | Readjust the entire document. |

# 1. Introduction

Welcome to the Supervyse OPF Web UI Developer's Guide! This guide is designed to help you navigate the process of creating, customizing, and optimizing web interfaces for the Supervyse OPF. Whether you are building a new web page, customizing the web logo, adjusting the menu style, or managing the visibility of pages, this guide will provide you with the knowledge and tools to get the job done efficiently.

In this guide, you will learn how to:

- **Create a New Web Page**: Step through the process of creating new web pages within the Supervyse OPF environment, including how to structure the page, add elements, and integrate backend functionality.
- **Toggle Page Visibility:** Learn how to manage page visibility by toggling pages on or off. This feature allows you to control which pages are supported on different platforms, enabling flexible page management.
- **Customize the Web Logo**: Learn how to modify or replace the default web logo to reflect your branding, with a focus on responsive design to ensure it looks great on any device or screen size.
- **Customize Menu Style**: Discover how to adjust the style of the navigation menu to align with your design preferences, including changing colors, fonts, and layout, or creating entirely new menu structures.
- **Developing with the Vue Dev Server**: Understand how to use the Vue Dev Server for local development, including running and testing your web pages in a live environment with features such as hot reloading and proxying to backend services.

This guide assumes a foundational understanding of web development, including **HTML**, **CSS**, **JavaScript**, and the **Vue.js** framework. By the end of this guide, you will have the skills needed to enhance the Supervyse OPF Web UI, create web pages, and deliver a unique user experience tailored to your organization’s needs.

Let’s get started on creating powerful, personalized web interfaces with Supervyse OPF!

# 2. Customize a New Web Page

This section provides a guide on how to create a new web page and integrate it into the web menu using the Vue framework. It also covers how to toggle page visibility, offering you flexible control over page accessibility and efficient page management.

## 2.1 Frontend

To add a new page in the frontend, follow these steps:

- Add menu items.
- Create and integrate new page components.
- Include relevant language strings.
- Add necessary stores for state management.

### 2.1.1 Add Menu Items

Modify the following file and add menu-related settings.
Location: isc-src/web/webui-vue/src/env/insyde/json/menu.json
Insert the following code under "logs" -> "children" section (see preview below).

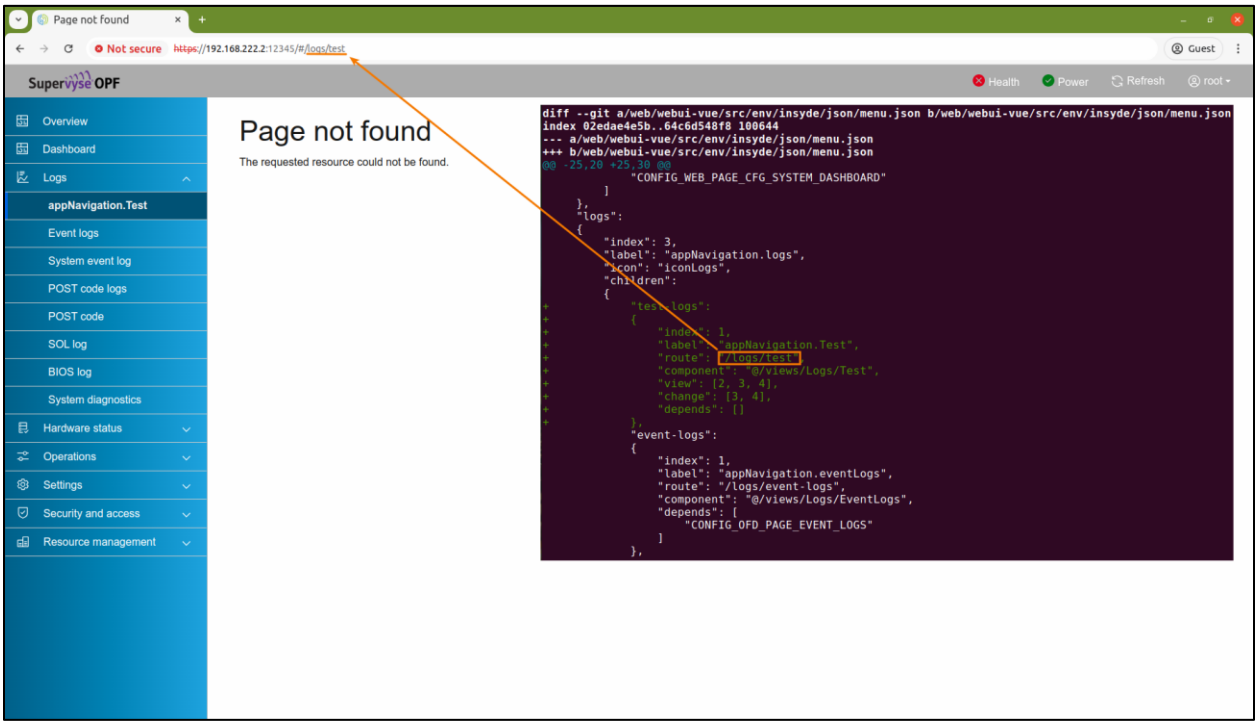
```
diff --git a/web/webui-vue/src/env/insyde/json/menu.json b/web/webui-vue/src/env/insyde/json/menu.json
index 02edae4e5b..64c6d548f8 100644
--- a/web/webui-vue/src/env/insyde/json/menu.json
+++ b/web/webui-vue/src/env/insyde/json/menu.json
@@ -25,20 +25,30 @@
            "CONFIG_WEB_PAGE_CFG_SYSTEM_DASHBOARD"
        ]
    },
    "logs":
    {
        "index": 3,
        "label": "appNavigation.logs",
        "icon": "iconLogs",
        "children":
        {
+           "test-logs":
+           {
+               "index": 1,
+               "label": "appNavigation.Test",
+               "route": "/logs/test",
+               "component": "@/views/Logs/Test",
+               "view": [2, 3, 4],
+               "change": [3, 4],
+               "depends": []
+           },
            "event-logs":
            {
                "index": 1,
                "label": "appNavigation.eventLogs",
                "route": "/logs/event-logs",
                "component": "@/views/Logs/EventLogs",
                "depends": [
                    "CONFIG_OFD_PAGE_EVENT_LOGS"
                ]
            },
            "test-logs":
            {
                "index": 1,
                "label": "appNavigation.Test",
                "route": "/logs/test",
                "component": "@/views/Logs/Test",
```

```
                "view": [2, 3, 4],
                "change": [3, 4],
                "depends": []
            },
```

**Field Description**:

- ***test-logs***: A unique identifier for this element, which can be any non-repeating string compared to sibling elements.
- **index**: Defines the menu order (items with lower numbers appear first; items with the same number are displayed in the order they are listed).
- **label**: Specifies the menu name and page title.
- **route**: Displays the path in the URL.
- **component**: The location of the page component.
  *Note: The '@' symbol corresponds to the 'src' path and must be used as '@' here.*
- **view**: Permission to view the page.
  Roles associated with this permission: 4: Administrator, 3: Operator, 2: ReadOnly.
- **change**: Permission to modify the page.
  Roles associated with this permission: 4: Administrator, 3: Operator, 2: ReadOnly.
- **disable**: Page Visibility: The default value is false.
  (When set to true, the page will be hidden.)
- **depends**: Dependency on Settings: The page will be hidden if the necessary settings are not met. This feature enables toggling page visibility through menuconfig, providing greater control over which pages are accessible.

The preview is shown in the image below. Since the page component has not been implemented yet, a 'page not found' message will be displayed.




## 2.1.2 Create and Integrate a New Page Component

This section will guide you through creating a new page, Test.vue, that includes a message and two buttons. The message content will be updated upon clicking either of the buttons.

The following is an example of how to modify it.

Location: isc-src/web/webui-vue/src/views/Logs/Test/index.js

```
import Test from './Test.vue';
export default Test;
```

Location: isc-src/web/webui-vue/src/views/Logs/Test/Test.vue

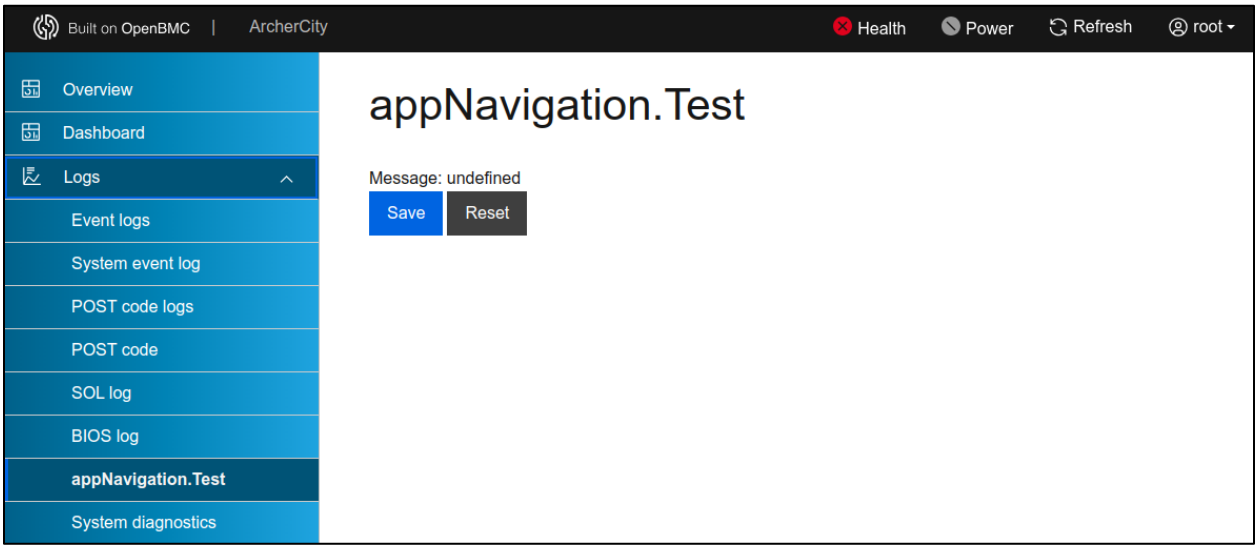
```
<template>
  <b-container fluid="xl">
    <page-title />
    <b-row>
      <b-col>
        <div>{{ `${title}: ${message}` }}</div>
      </b-col>
    </b-row>
    <b-row>
      <b-col>
        <b-button variant="primary" class="mr-1" @click="onSave">
          {{ $t('global.action.save') }}
        </b-button>
        <b-button variant="secondary" class="mr-1" @click="onReset">
          {{ $t('global.action.reset') }}
        </b-button>
      </b-col>
    </b-row>
  </b-container>
</template>

<script>
import PageTitle from '@/components/Global/PageTitle';

export default {
  components: {
    PageTitle,
  },
  data() {
    return {
      title: 'Message',
    };
  },
  computed: {
    message() {
      return this.$store.getters['mytest/message'];
    },
  },
  created() {
    this.startLoader();
    // Do something
    this.endLoader();
  },
  methods: {
    onSave() {
      this.startLoader();
      this.$store
        .dispatch('mytest/save')
        .catch(({ message }) => this.errorToast(message))
        .finally(() => this.endLoader());
    },
    onReset() {
```

```
      this.startLoader();
      this.$store
        .dispatch('mytest/reset')
        .catch(({ message }) => this.errorToast(message))
        .finally(() => this.endLoader());
    },
  },
};
</script>
```

Preview:




### 2.1.3 Include Relevant Language Strings

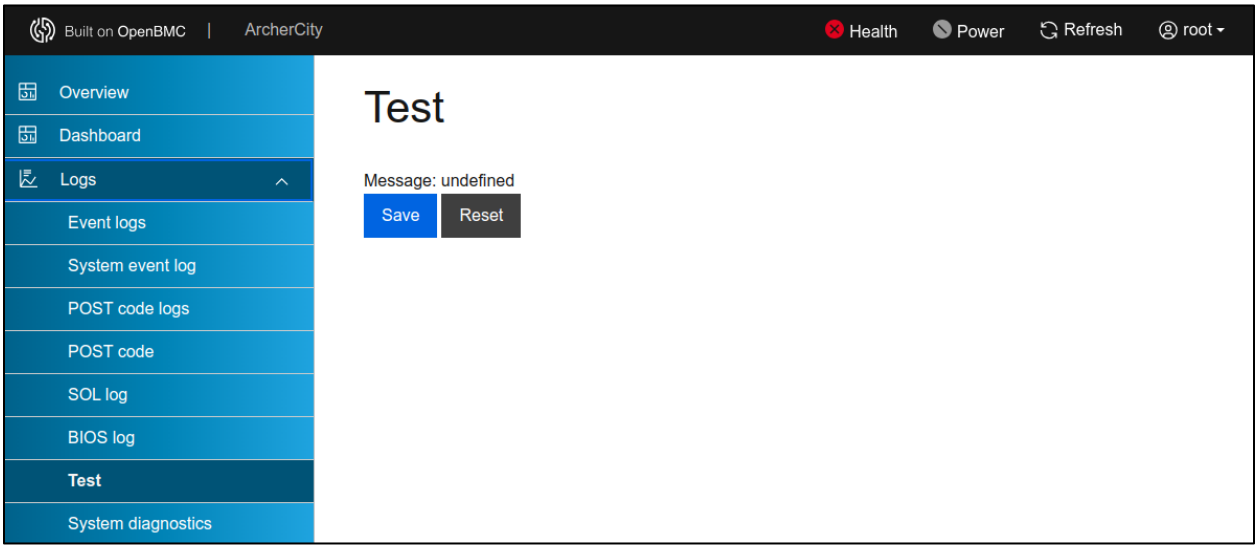
Add language strings to support multiple languages, defining keys and corresponding strings.

| Path | Description |
|---|---|
| isc-src/web/webui-vue/src/env/insyde/locale/en-US.json | English string file. |
| isc-src/web/webui-vue/src/env/insyde/locale/zh-CN.json | Simplified Chinese string file. |
| isc-src/web/webui-vue/src/env/insyde/locale/zh-TW.json | Traditional Chinese string file. |

Example: isc-src/web/webui-vue/src/env/insyde/locale/en-US.json

```
diff --git a/web/webui-vue/src/env/insyde/locales/en-US.json b/web/webui-vue/src/env/insyde/locales/en-US.json
index f680471b57..989bb5507b 100644
--- a/web/webui-vue/src/env/insyde/locales/en-US.json
+++ b/web/webui-vue/src/env/insyde/locales/en-US.json
@@ -229,6 +229,7 @@
     "titleRefresh": "Refresh application data"
   },
   "appNavigation": {
+    "Test": "@:appPageTitle.Test",
     "snmp": "@:appPageTitle.snmp",
     "tacacs": "@:appPageTitle.tacacs",
     "virtualFrontPanel": "@:appPageTitle.virtualFrontPanel",
@@ -305,6 +306,7 @@
     "powerStatistics": "@:appPageTitle.powerStatistics"
   },
   "appPageTitle": {
+    "Test": "Test",
     "snmp": "SNMP",
     "tacacs": "TACACS+",
     "virtualFrontPanel": "Virtual front panel",
```

Preview:




### 2.1.4 Add Necessary Stores for State Management

In the Vue framework, particularly when using Vuex (the state management library for Vue), a "store" is a centralized repository for managing the state of your application.

Add a new store<TestStore> and named it mytest.
Location: isc-src/web/webui-vue/src/store/index.js

```
diff --git a/web/webui-vue/src/store/index.js b/web/webui-vue/src/store/index.js
index d7c1b22d03..adbb82ae7d 100644
--- a/web/webui-vue/src/store/index.js
+++ b/web/webui-vue/src/store/index.js
@@ -25,6 +25,7 @@ import BmcStore from './modules/HardwareStatus/BmcStore';
 import ProcessorStore from './modules/HardwareStatus/ProcessorStore';
 import AssemblyStore from './modules/HardwareStatus/AssemblyStore';
 import PostCodeLogsStore from './modules/Logs/PostCodeLogsStore';
+import TestStore from './modules/Logs/TestStore';
 import PoliciesStore from './modules/SecurityAndAccess/PoliciesStore';
 import FactoryResetStore from './modules/Operations/FactoryResetStore';

@@ -64,6 +65,7 @@ export default new Vuex.Store({
     processors: ProcessorStore,
     assemblies: AssemblyStore,
     postCodeLogs: PostCodeLogsStore,
+    mytest: TestStore,
     virtualMedia: VirtualMediaStore,
     policies: PoliciesStore,
     factoryReset: FactoryResetStore,
```

Location: isc-src/web/webui-vue/src/store/modules/Logs/TestStore.js

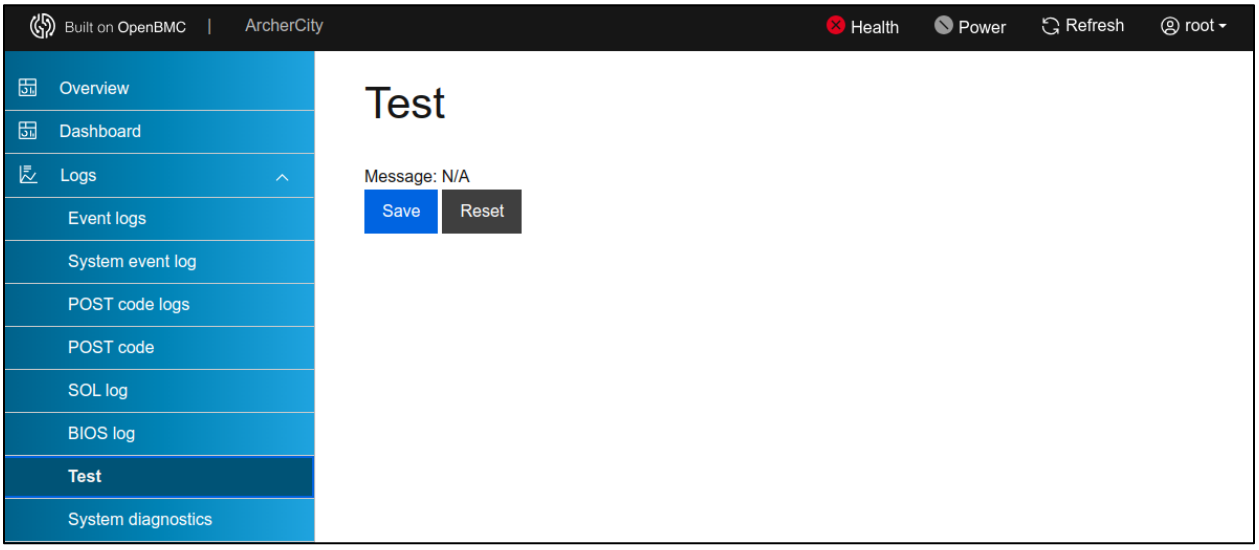
```
import api from '@/store/api';

const TestStore = {
  namespaced: true,
  state: {
    message: 'N/A',
  },
```

```
  getters: {
    message: (state) => state.message,
  },
  mutations: {
    setMessage: (state, msg) => {
      state.message = msg;
    },
  },
  actions: {
    async save({ commit }) {
      return await api
        .get('/cgi/mypage.cgi')
        .then(({ data = {} }) => commit('setMessage', data.POWER_STATUS))
        .catch((error) => {
          console.log(error);
          throw new Error('Save error', error);
        });
    },
    async reset({ commit }) {
      return await api
        .get('/redfish/v1/Systems/system')
        .then(({ data = {} }) => commit('setMessage', data.Model))
        .catch((error) => {
          console.log(error);
          throw new Error('Reset error', error);
        });
    },
  },
};

export default TestStore;
```

Preview:



## 2.2 Backend

A Backend Web API (Application Programming Interface) is a server-side application or set of services that handle requests from clients, typically web or mobile applications. Web APIs provide data or functionality that can be accessed over HTTP(S) through endpoints (URLs), allowing various front-end interfaces (such as Vue applications) to interact with the backend for data or actions.

The following sections will teach you how to add a new CGI Web API.

### 2.2.1 CGI Source Code

The following example will add a new Web API to retrieve the current power status.

URL: https://xxx.xxx.xxx.xxx/cgi/mypage.cgi
Method: GET
Expected result:

```
{
  "POWER_STATUS": "off"
}
```

Location: isc-src/web/cgi/mypage.c

```
#include <utils.h>
#include <mci_chassis.h>

static void get_data(cgi_params *cpsp)
{
    int ret = 0;
    int status = 0;
    int data_length = 0;

    ret = mci_get_chassis_status(&status, NULL, &data_length);
    if (ret == 0) {
        cJSON *json = cJSON_CreateObject();
        if (json) {
            cJSON_AddStringToObject(json, "POWER_STATUS", status == 1 ? "on" : "off");
            exit_json(cpsp, json);
        }
        exit_error(cpsp, 500, "cJSON_CreateObject fail");
    }
    exit_error(cpsp, 500, "mci_get_chassis_status fail");
}

int main(void)
{
    HANDLER handler_table[] = {
        /* method, query, privilege, handler_function */
        { "GET",  NULL, 2, get_data },
    };

    PROCESS_REQUEST(handler_table);
    return 0;
}
```

### 2.2.2 Build

```
# Perform a module clean for CGI.
```

```
bitbake cgi -c clean

# Perform a module compile for CGI.
bitbake cgi
```

After the module is built, libcgiapi.so will be updated at the location isc-src/web/cgi/oe-workdir/package/usr/lib/libcgiapi.so

### 2.2.3 Deploy

Replace libcgiapi.so on the BMC console and restart the bmcweb service to reload the .so file.

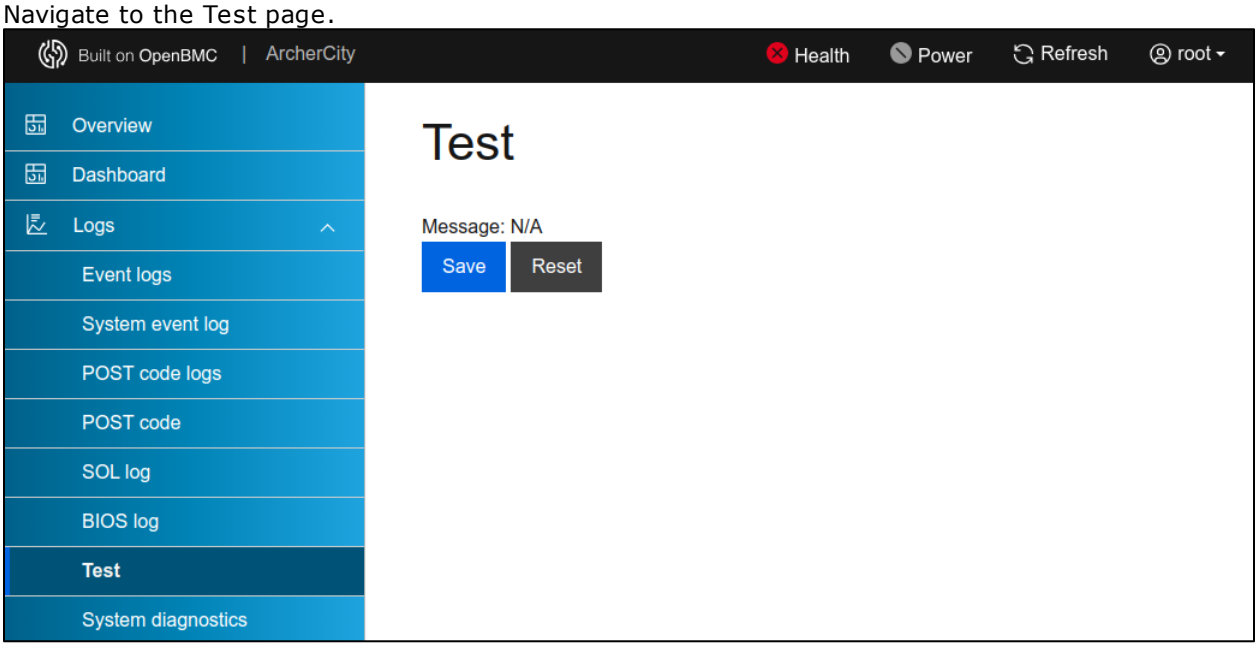
```
# <On the BMC console>
# Copy the new libcgiapi.so to BMC.
scp user@xxx.xxx.xxx.xxx:/yourSDK/isc-src/web/cgi/oe-workdir/package/usr/lib/libcgiapi.so
/usr/lib/

# Restart bmcweb service.
systemctl restart bmcweb
```

## 2.3 Verify

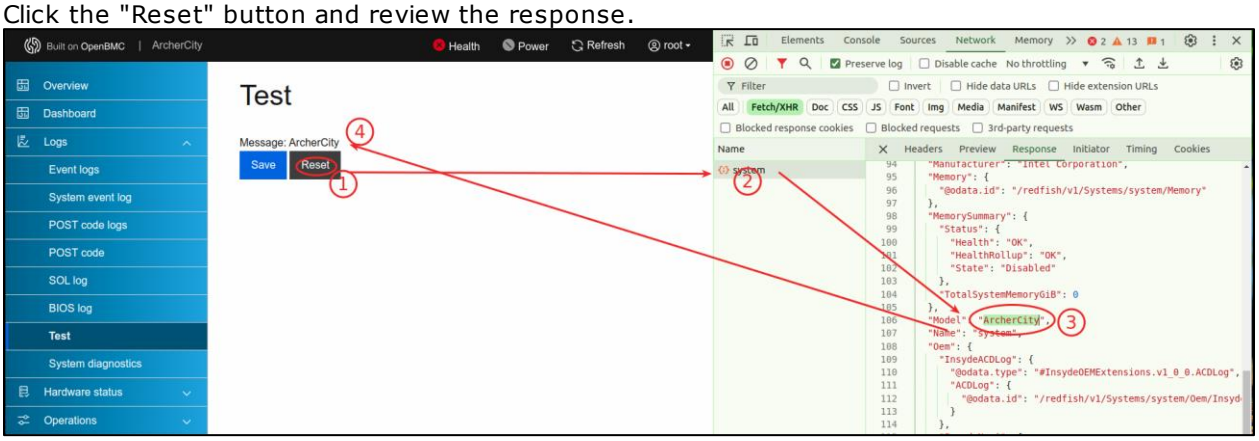
This section guides you on how to validate your modifications.

Navigate to the Test page.

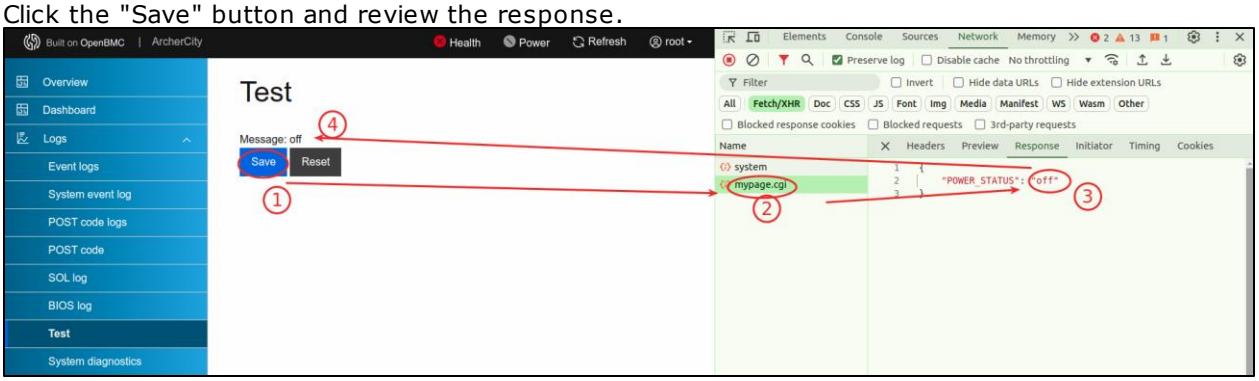



Press F12 on your keyboard to open the browser's "Developer Tools".

Click the "Reset" button and review the response.




Click the "Save" button and review the response.

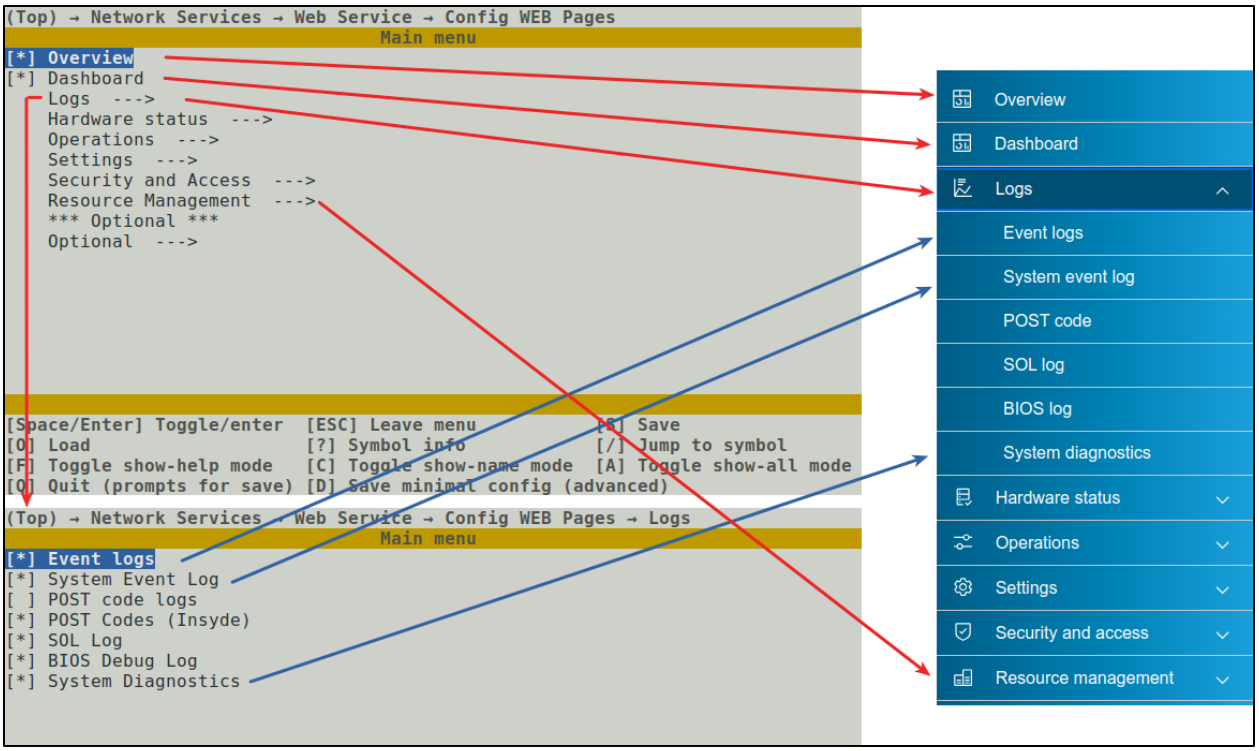



## 2.4 Customize Page Visibility

The visibility of individual pages can be managed through menuconfig. Here, this only controls the page visibility. Whether the page functions correctly depends on platform compatibility and underlying support.

**Steps:**

1. Run the command: bitbake pkconfig -c menuconfig
2. Select the corresponding pages to enable.

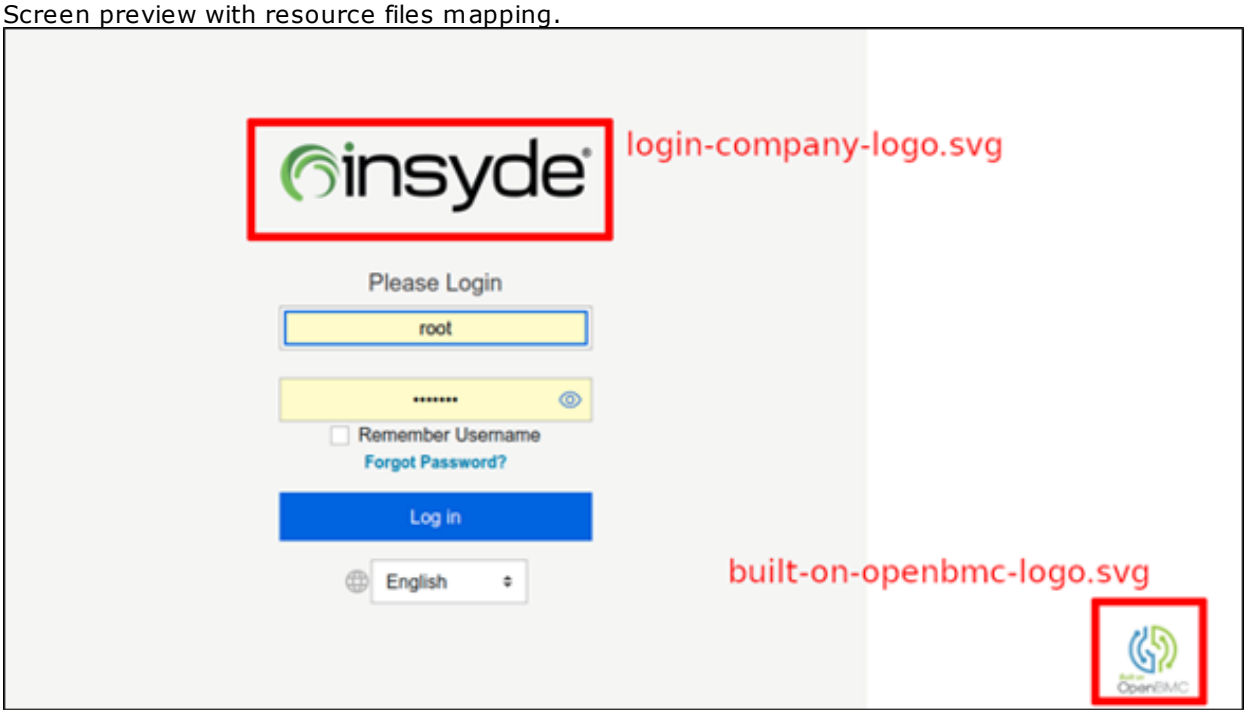

(Top) → Network Services → Web Service → Config WEB Pages
Main menu
[*] Overview
[*] Dashboard
Logs --->
Hardware status --->
Operations --->
Settings --->
Security and Access --->
Resource Management --->
*** Optional ***
Optional --->
[Space/Enter] Toggle/enter [ESC] Leave menu [S] Save
[O] Load [?] Symbol info [/] Jump to symbol
[F] Toggle show-help mode [C] Toggle show-name mode [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)
(Top) → Network Services → Web Service → Config WEB Pages → Logs
Main menu
[*] Event logs
[*] System Event Log
[ ] POST code logs
[*] POST Codes (Insyde)
[*] SOL Log
[*] BIOS Debug Log
[*] System Diagnostics
Overview
Dashboard
Logs
Event logs
System event log
POST code
SOL log
BIOS log
System diagnostics
Hardware status
Operations
Settings
Security and access
Resource management

# 3. Customize the Web Logo

This section offers a step-by-step guide on customizing the page logo.

Resource location: isc-src/web/webui-vue/src

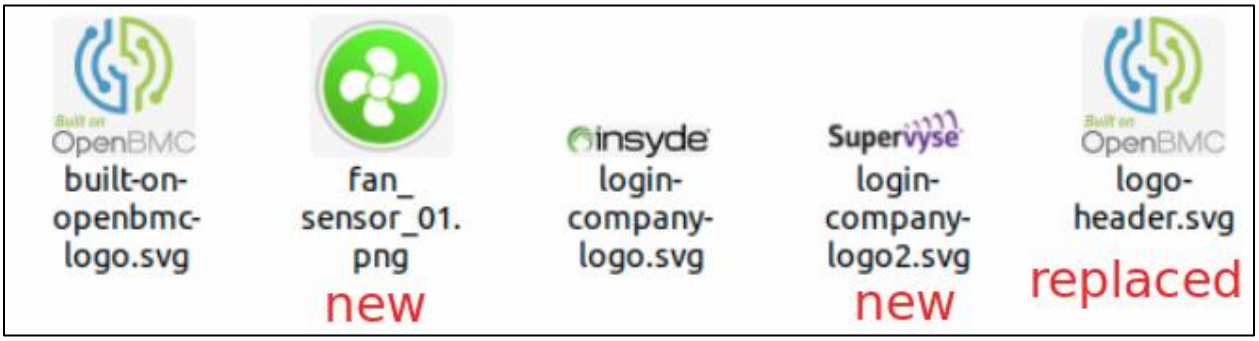
Screen preview with resource files mapping.



## 3.1 Replace the Logos

Update the logo and include additional images in the resource folder.

- Location: isc-src/web/webui-vue/src/assets/images
- Add two image files, fan_sensor_01.png and login-company-logo2.svg, to apply the new logo on the login page.
- Replace logo-header.svg with your company's logo.

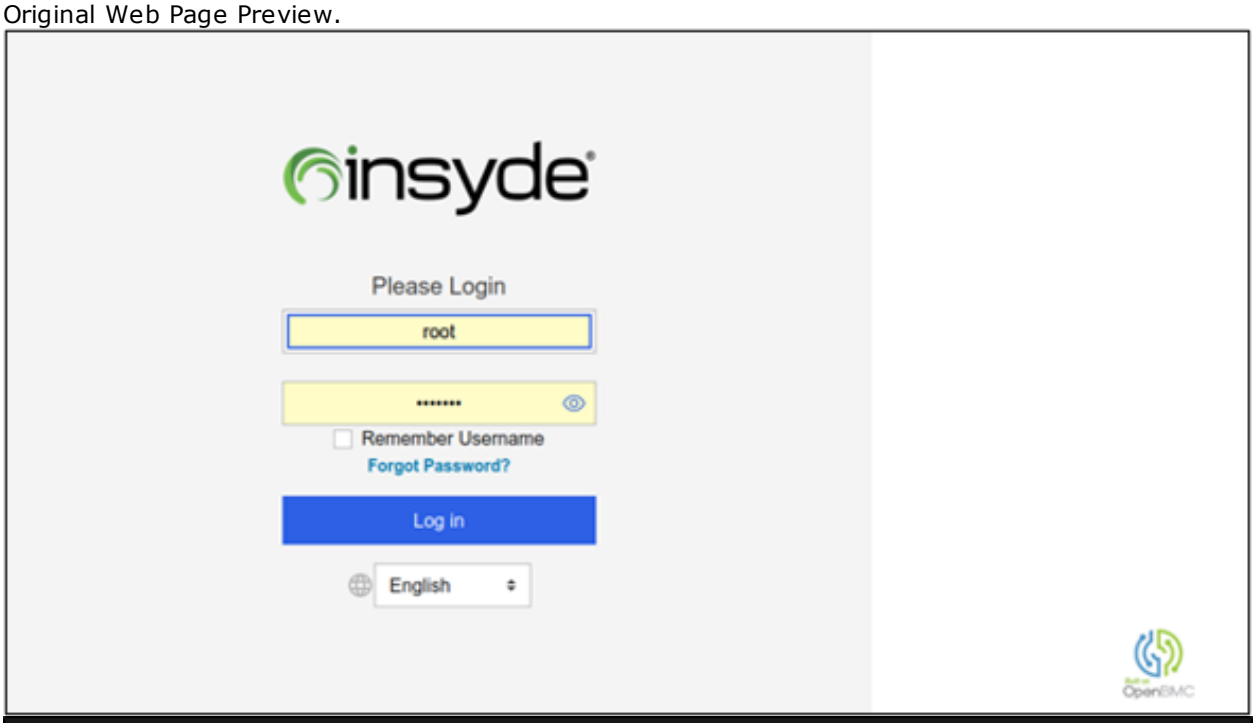


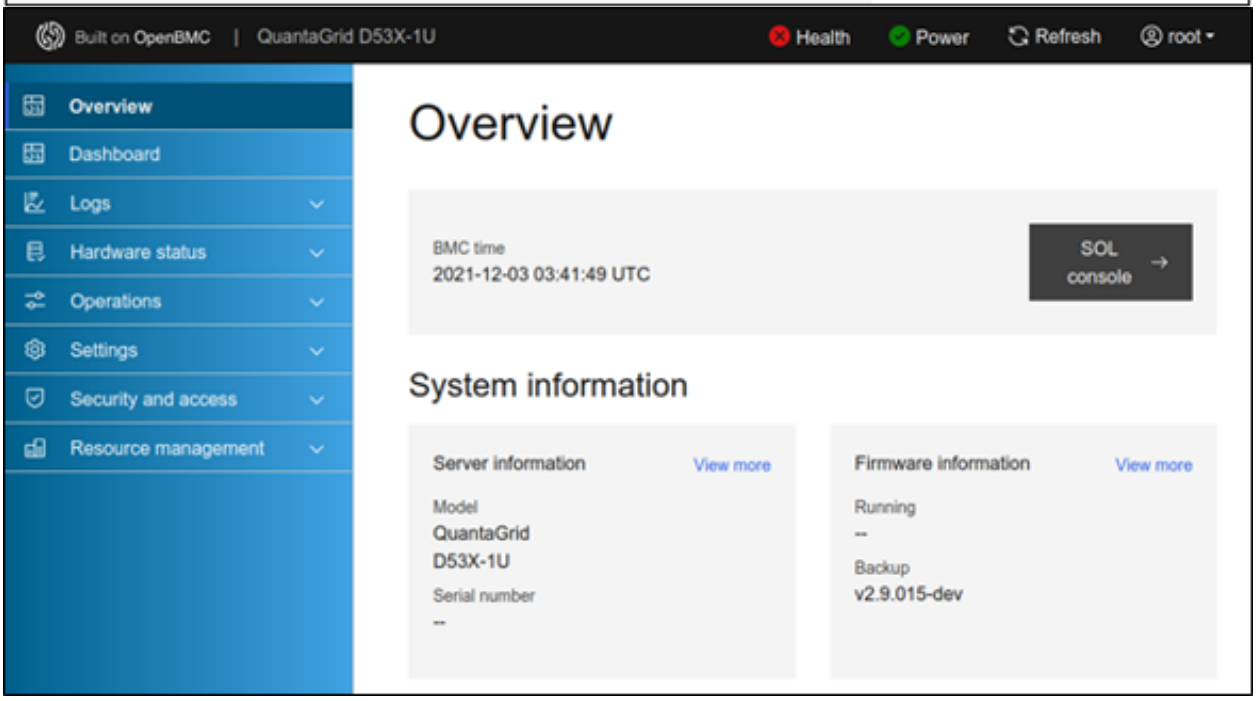

Modify the Vue file for the login page to update the logo.

- Location: isc-src/web/webui-vue/src/layouts/LoginLayout.vue
- Replace the company logo from login-company-logo.svg to login-company-logo2.svg.
- Add an icon to the left side of the OpenBMC logo.

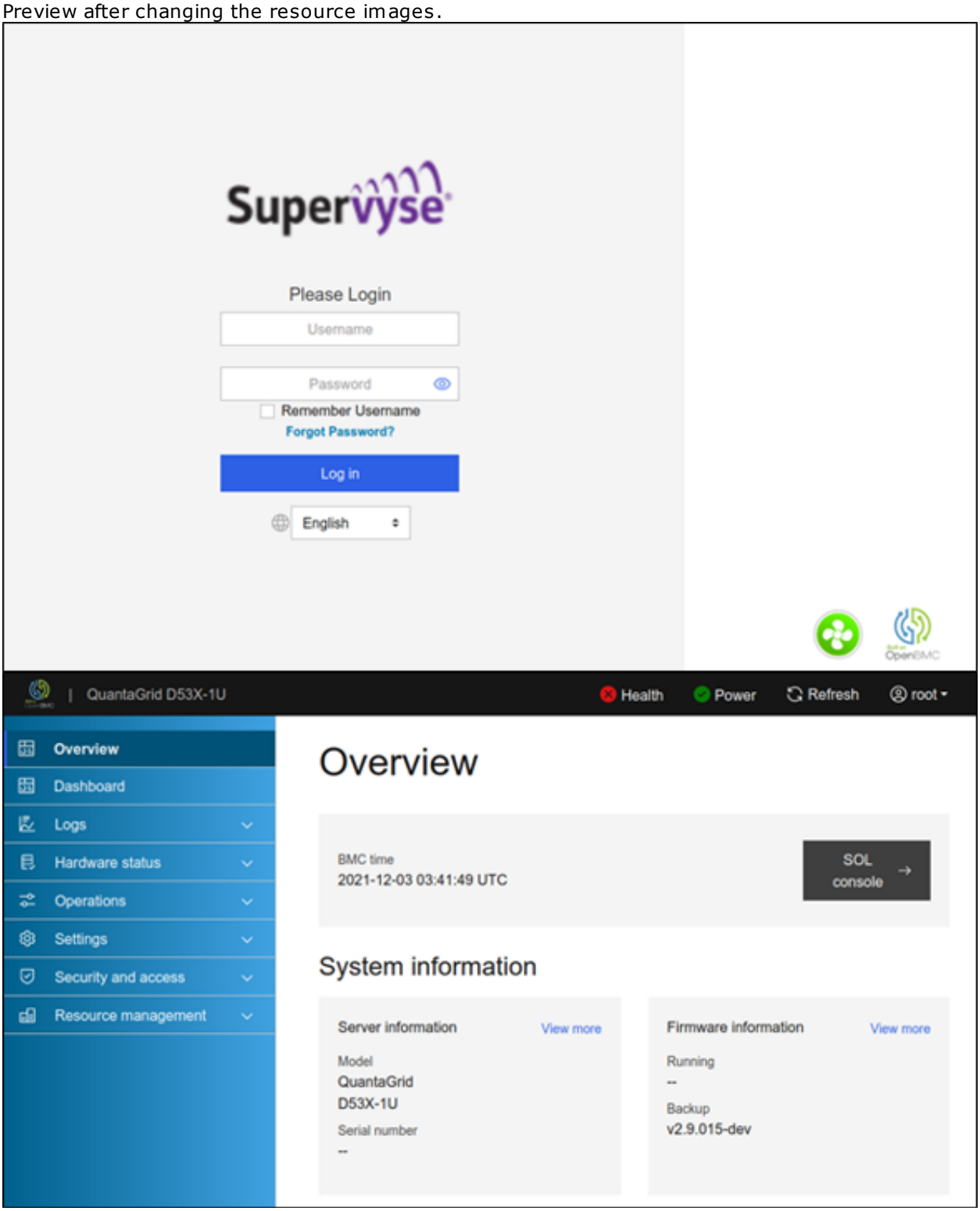
```
1   <template>
2     <main>
3       <div class="login-container">
4         <div class="login-main">
5           <div>
6             <div class="login-brand mb-5" @dblclick="handler">
7               <img
8                 width="270px"
9                 src="@/assets/images/login-company-logo2.svg"
10                :alt="altLogo"
11              />
12            </div>
13            <h1 v-if="customizableGuiName" class="h3 mb-5">
14              {{ customizableGuiName }}
15            </h1>
16            <router-view class="login=form form-background" />
17          </div>
18        </div>
19        <div class="login-aside">
20          <div class="login-aside__logo-brand">
21            <!-- Add Secondary brand logo if needed -->
22          </div>
23          <div class="login-aside__logo-bmc">
24            <img
25              height="60px"
26              src="@/assets/images/fan_sensor_01.png"
27              alt="Built on OpenBMC"
28            />
29          </div>
30          <div class="login-aside__logo-bmc">
31            <img
32              height="60px"
33              src="@/assets/images/built-on-openbmc-logo.svg"
34              alt="Built on OpenBMC"
35            />
36          </div>
37        </div>
38      </div>
39    </main>
40  </template>
41
42  <script>
```

Original Web Page Preview.



Preview after changing the resource images.

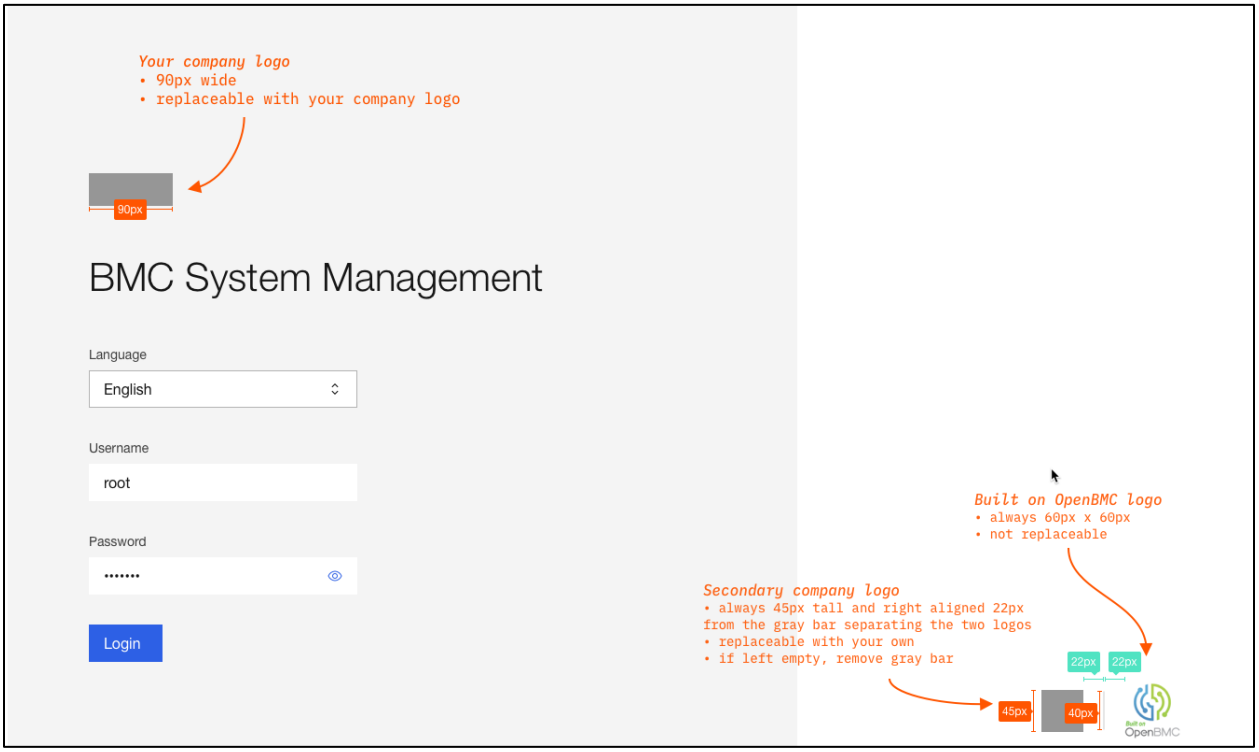


## 3.2 References

The customization guide for the logos on web pages is based on the OpenBMC guidance provided below:

1. https://github.com/openbmc/docs/blob/master/development/web-ui.md
2. https://github.com/openbmc/webui-vue/issues/63

The first page describes how to replace the login-company-logo image on the login page and the logo-header image on every page after login. The second page describes how to put your company logo alongside the built-on-openbmc-logo image on the bottom right corner of the login page. It also suggests that the built-on-openbmc-logo image is not replaceable.

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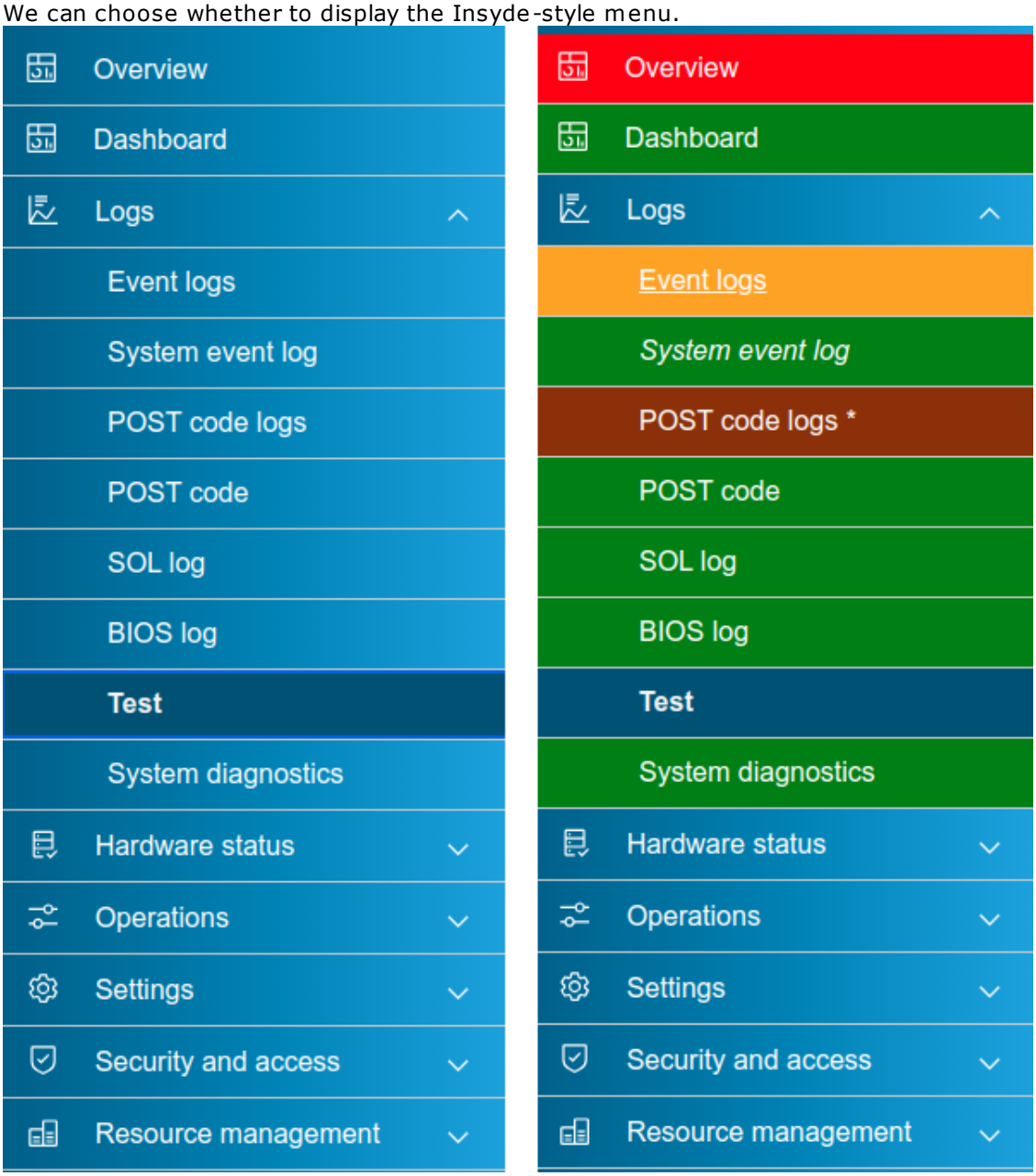
# 4. Customize Menu Style

How do I customize the style of a menu item? Something like this.
We can choose whether to display the Insyde-style menu.



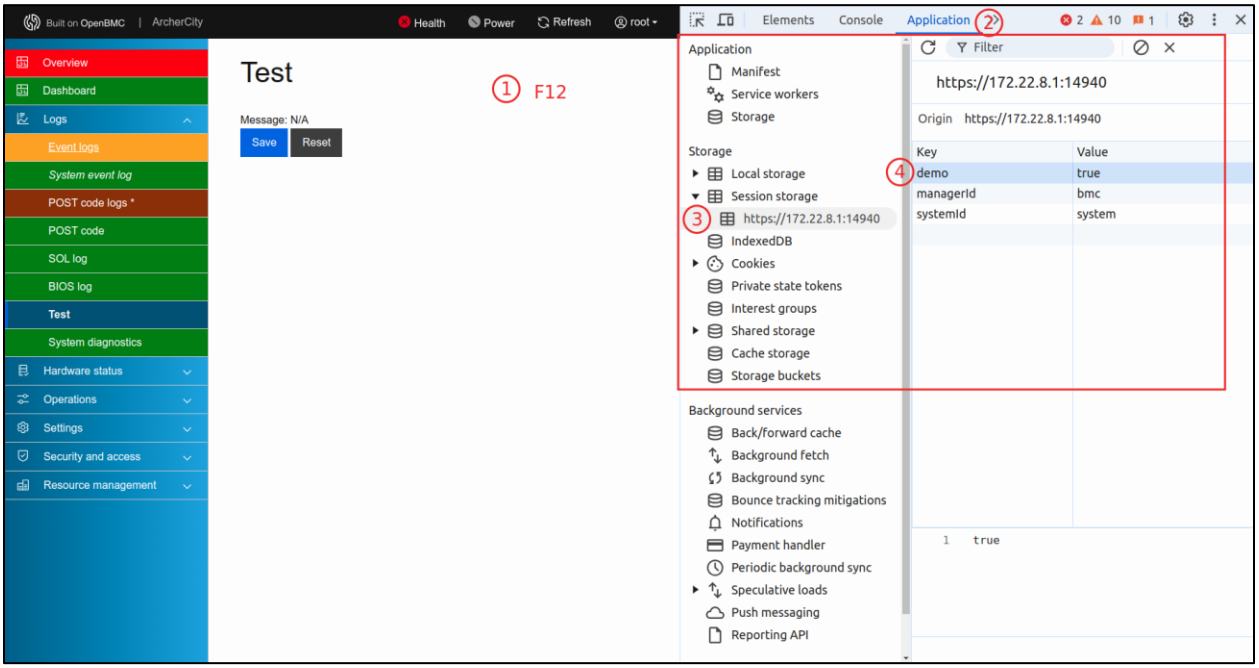

## 4.1 Enable Menu Style

First, enable Demo mode, as the menu style is only functional in this mode.

Location: isc-src/web/webui-vue/.env.insyde
Add the following setting to enable it.

```
VUE_APP_DEMO=true
```

=true (default on)
=false or SKIP (default off)
The default value of demo is stored in Session Storage.




1. Press F12 on your keyboard to open the browser's "Developer Tools".

2. Select the "Application" tab.

3. Choose "Session storage" from the left-hand menu.

4. Variables are listed in the key/value format on the right side.
   Toggle the demo variable (true or false) and press F5 (browser reload) to apply.
   This value will reset to default when the page is closed and reopened.

## 4.2 Configure Menu Style

This section provides guidance on how to customize the style of the menu.

Location: isc-src/web/webui-vue/src/env/insyde/json/menu.json
Add or modify the "myclass" for a menu item to customize its style.

```
diff --git a/web/webui-vue/src/env/insyde/json/menu.json b/web/webui-vue/src/env/insyde/json/menu.json
index 6c24e3b0db..6dee3fa84f 100644
--- a/web/webui-vue/src/env/insyde/json/menu.json
+++ b/web/webui-vue/src/env/insyde/json/menu.json
@@ -2,6 +2,7 @@
     "overview":
     {
         "index": 1,
+        "myclass": "red",
         "label": "appNavigation.overview",
         "route": "/",
         "component": "@/views/Overview",
@@ -35,6 +36,7 @@
             "event-logs":
             {
                 "index": 1,
+                "myclass": "orange underline",
                 "label": "appNavigation.eventLogs",
                 "route": "/logs/event-logs",
                 "component": "@/views/Logs/EventLogs",
@@ -45,7 +47,7 @@
             "sel":
             {
                 "index": 2,
-                "myclass": "green",
+                "myclass": "green italic",
                 "label": "appNavigation.sel",
                 "route": "/logs/sel",
                 "component": "@/env/insyde/views/SEL",
@@ -58,6 +60,7 @@
             "test-logs":
             {
                 "index": 7,
+                "myclass": "rgb(255, 0, 255)",
                 "label": "appNavigation.Test",
                 "route": "/logs/test",
                 "component": "@/views/Logs/Test",
@@ -68,6 +71,7 @@
             "post-code-logs":
             {
                 "index": 3,
+                "myclass": "#883300 insyde",
                 "label": "appNavigation.postCodeLogs",
                 "route": "/logs/post-code-logs",
                 "component": "@/views/Logs/PostCodeLogs",
```

How do you utilize the "myclass" property to customize the menu item's appearance?
Example: "myclass": "insyde italic bold underline color".

- insyde: Add the "insyde" class to include additional content (*) and apply the Insyde-selected back-groupd color.
- italic: Font style: italic.
- bold: Font style: bold. (Selected items are bold by default and are not prominent when using this style.)
- underline: Font style: underline.
- *color*: Supported formats: *red* (color_name), *#ff0000*, or rgb(255, 0, 0).

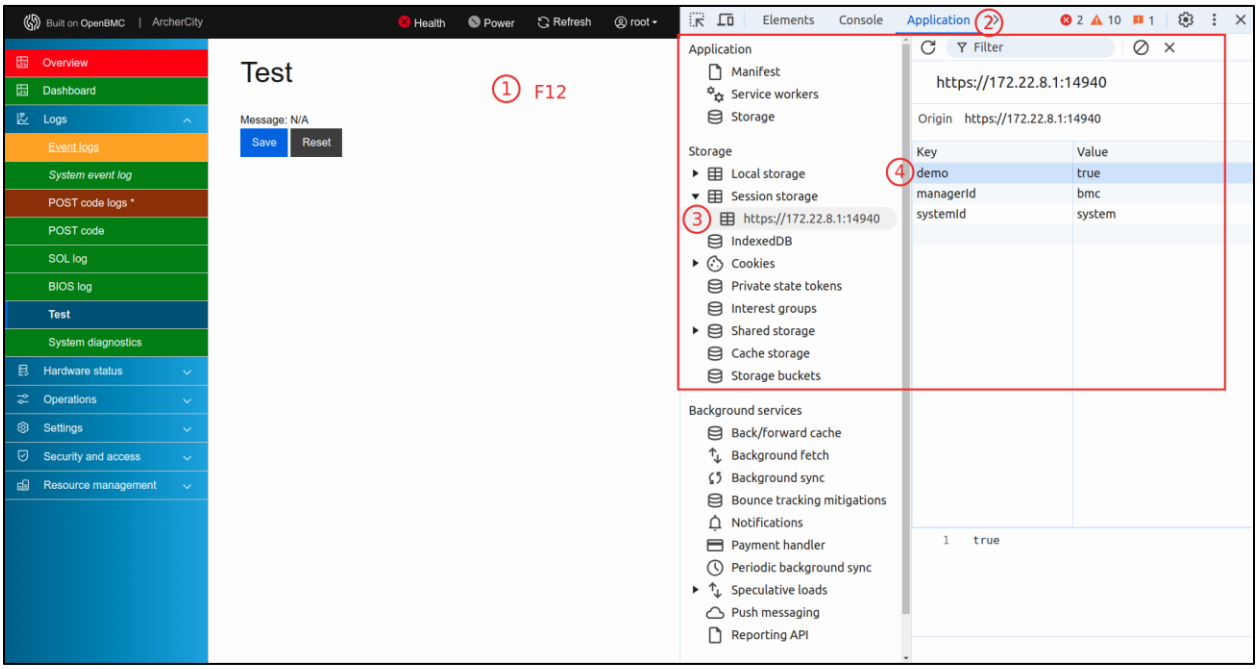
User-defined insyde class.

- Location: isc-src/web/webui-vue/src/components/AppNavigation/AppNavigation.vue
- When you define "insyde" in myclass, a * will be appended at the end.
- You can change this appended value here.

```
181  .insyde a::after,
182  a.insyde::after,
183  button.insyde::after {
184    content: ' *';
185  }
```

## 4.3 Enable Menu Style for One-Time Use at Runtime

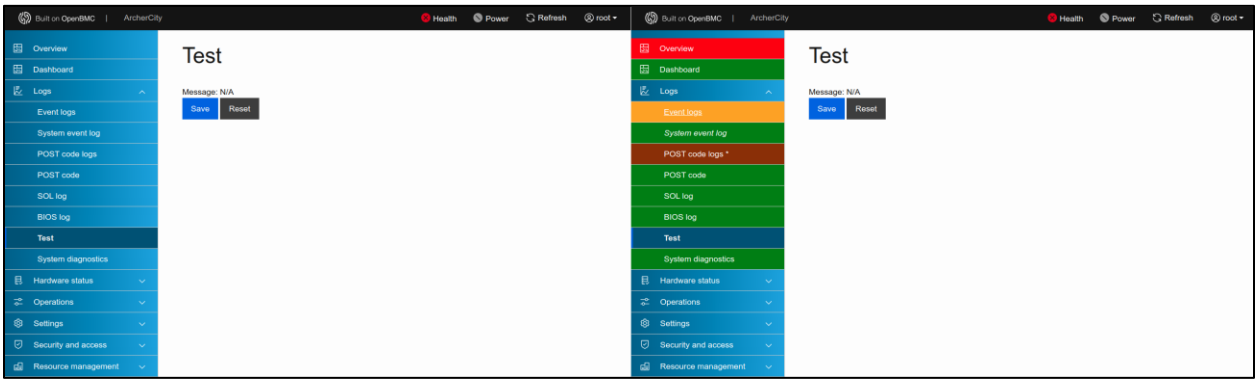
If you don't want to recompile the program and just want to see the effect, you can modify the demo value stored in Session Storage.




1. Press F12 on your keyboard to open the browser's "Developer Tools".
2. Select the "Application" tab.
3. Choose "Session storage" from the left-hand menu.
4. Variables are listed in the key/value format on the right side.
   Toggle the demo variable (true or false) and press F5 (browser reload) to apply.
   This value will reset to default when the page is closed and reopened.

## 4.4 Enable Normal/Demo Mode Toggling

To switch between the normal mode (left) and demo mode (right) menu styles without recompiling the code, you can use the following method: double-click the Insyde logo on the login page to toggle between the two modes.




To enable this feature, first enable the corresponding option in menuconfig, save your changes, and then proceed with the image build process. After completing these steps, you will be able to switch modes at any time without the need to recompile.

Steps to enable it:

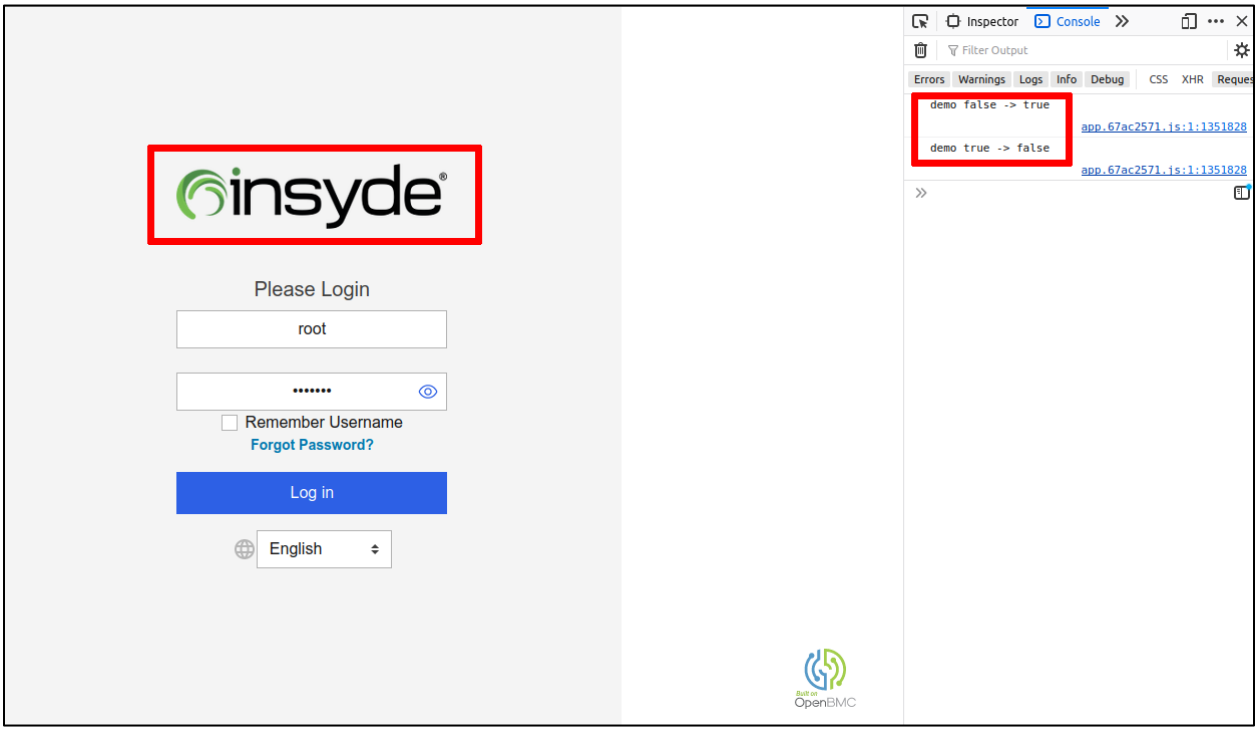
1. Run the command: bitbake pkconfig -c menuconfig
2. Select the corresponding option in the menuconfig.

```
(Top) → Network Services → Web Service → Web parameters
                                Main menu
(128) User name max length on BMC web
(128) Password max length on BMC web
(180) Web heartbeat timeout
(10) Web heartbeat interval
(0) Timeout time of IKVM session connection
(30) Timeout time of IKVM socket connection
[ ] BMC Setting Preserve Option
[ ] Web page demo toggle

[Space/Enter] Toggle/enter  [ESC] Leave menu          [S] Save
[O] Load                    [?] Symbol info           [/] Jump to symbol
[F] Toggle show-help mode   [C] Toggle show-name mode  [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)
```

After building the image and updating it to the BMC, double-click the Insyde logo on the login page to toggle to demo mode.

Check the demo mode flags by using the browser's developer console.



# 5. Developing with the Vue Dev Server

The **Vue dev server** refers to the development server used by Vue.js applications. When you set up a Vue project, the development server is automatically included as part of the Vue CLI (Command Line Interface). It serves your application, provides live reloading, and allows for quick development iteration.

Key Features of Vue Dev Server:

- **Hot Module Replacement (HMR)**: Updates the modules without a full page reload.
- **Live Reload**: Automatically reloads the page when changes are detected.
- **Proxy Support**: Useful for handling API requests while developing locally.
- **Customizable**: Port, proxy, and other settings can be configured in the vue.config.js file.

## 5.1 Configuring Vue Dev Server

Create File: isc-src/web/webui-vue/.env.insyde.local

```
NODE_ENV=developement
BASE_URL='https://192.168.22.156:443'
VUE_APP_PORT=12345
VUE_APP_LOG_LEVEL='debug'
```

**Field Description:**

- **NODE_ENV**: Set this to development to enable development mode.
- **BASE_URL**: Set the proxy URL to forward frontend requests to the backend server.
- **VUE_APP_PORT**: Set the port for the development server to run on. Default is 8000.
- **VUE_APP_LOG_LEVEL**: Set the level of detail for the log output. Available options include silent, error, warn, info, and debug.

## 5.2 Running the Vue Dev Server

First, navigate to the isc-src/web/webui-vue directory.

Next, execute the following command:

```
npm install
npm run serve:insyde
```

Here's an explanation of the two commands:

- **npm install**
  This command is used to install all the dependencies listed in your project's package.json file. When you run this command, npm (Node Package Manager) will:
  - Download all the required packages and libraries that your project depends on.
  - Create a node_modules directory where these packages will be stored.
  - Create a package-lock.json (if it doesn't already exist), which locks the versions of the installed packages to ensure consistency across different environments.
- **npm run serve:insyde**
  This is a custom command defined in the scripts section of the package.json file of your project. The npm run command allows you to run any custom scripts that are defined in the scripts section. In this case, serve:insyde refers to a development server configuration specifically tailored for Insyde-related settings or features.

**In Summary:**

- **npm install**: Installs all the dependencies needed for your project to run.
- **npm run serve:insyde**: Runs the development server with a custom configuration, likely related to Insyde-specific features.

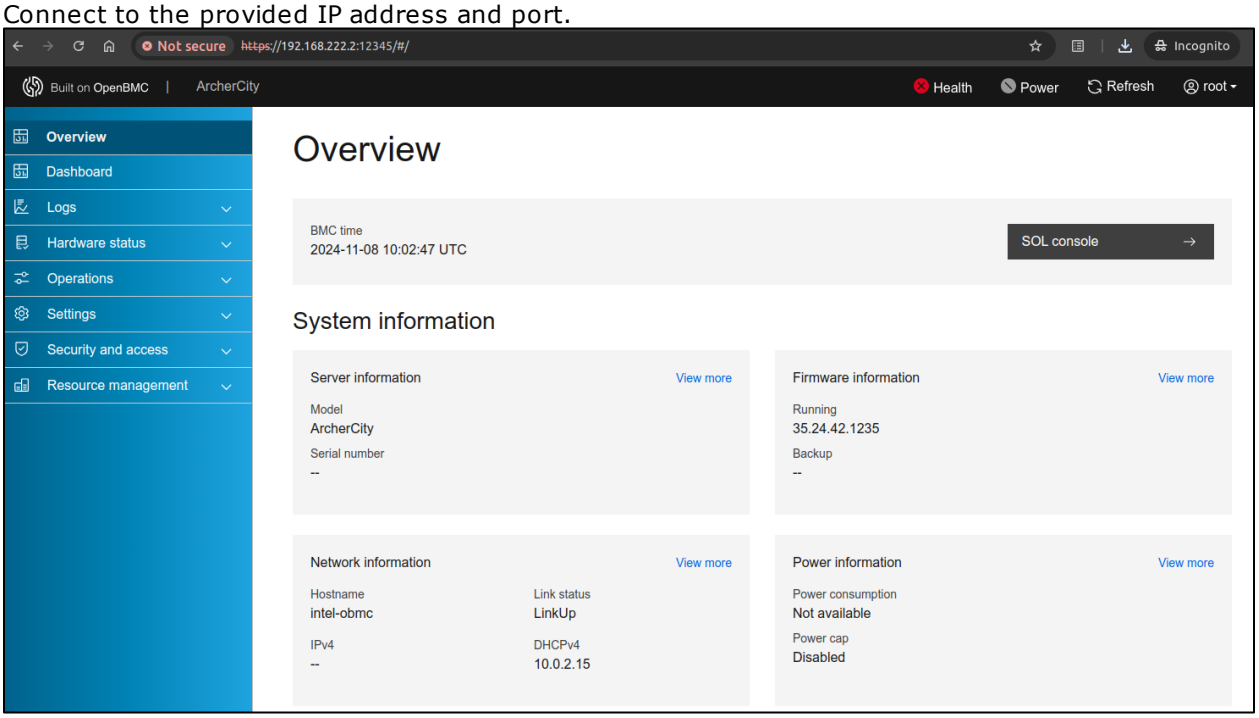
The screen displayed after startup is shown below.

```
DONE  Compiled successfully in 10902ms


App running at:
- Local:   https://localhost:12345/
- Network: https://192.168.222.2:12345/

Note that the development build is not optimized.
To create a production build, run npm run build.
```

Connect to the provided IP address and port.



# 6. Vue with ESLint

Vue projects frequently incorporate ESLint to maintain code quality and adhere to best practices. ESLint serves as a powerful linter that checks JavaScript code, enforcing style rules, catching syntax issues, and helping to prevent bugs.

**Key Benefits of Using ESLint with Vue**

- **Enforces Coding Standards**: ESLint applies a set of predefined rules to ensure code follows team or industry standards.
- **Error Prevention**: Identifies syntax and logical errors, helping prevent common bugs.
- **Vue-Specific Rules**: With eslint-plugin-vue, ESLint provides Vue-specific rules that optimize .vue files, including templates, styles, and script sections.

By integrating ESLint with Vue, teams gain better control over code quality, making the development process smoother and the code easier to maintain.

## 6.1 How to Use ESLint Verify

In the ESLint CLI, you can use the eslint --fix command to check and automatically fix code issues, or eslint --no-fix to only validate without modifying code.

First, navigate to the isc-src/web/webui-vue directory and make sure you have installed the node_modules by running the command:

```
npm install
```

Next, execute the following command to check or fix the related code:

```
npx eslint <file.js|file.vue>        # Checks if the code meets the defined standards
npx eslint <file.js|file.vue> --fix  # Checks and automatically corrects applicable issues
```

**Example:**

When you encounter the following error while compiling or running the Vue Dev Server:

```
 WARNING  Compiled with 1 warning

 warning  in ./src/views/Logs/Test/Test.vue

Module Error (from ./node_modules/eslint-loader/index.js):

/media/nvme4T/home/jamey/src/OPF-RV24.06.3m/isc-src/web/webui-vue/src/views/Logs/Test/Test.vue
  6:9  error  Delete `·`  prettier/prettier

✖ 1 problem (1 error, 0 warnings)
  1 error and 0 warnings potentially fixable with the `--fix` option.


 @ ./src/views sync .*\.(js|vue)$
 @ ./src/env/router/insyde.js
 @ ./src/router/index.js
 @ ./src/main.js
 @ multi (webpack)-dev-server/client?https://192.168.222.2:14940&sockPath=/sockjs-node (webpack)/hot/dev-server.js ./src/main.js
```

Run the following command to resolve it:

```
npx eslint src/views/Logs/Test/Test.vue --fix
```

This will automatically apply the necessary fixes to the specified file.

## 6.2 How to Config ESLint Rules

Location: isc-src/web/webui-vue/.eslintrc

**Configure ESLint Rules**: The .eslintrc file created will include rules for verifying code quality. You can modify the rules section to meet your project's coding standards. For example:

```
module.exports = {
  root: true,
  env: {
    node: true,
  },
  globals: {
    _: 'readonly',
  },
  extends: ['plugin:vue/recommended', 'eslint:recommended', '@vue/prettier'],
  rules: {
    'no-console': 'off',
    'no-debugger': process.env.NODE_ENV === 'production' ? 'error' : 'off',
    'prettier/prettier': [
      'error',
      {
        singleQuote: true,
      },
    ],
    'vue/component-name-in-template-casing': ['error', 'kebab-case'],
  },
  parserOptions: {
    parser: 'babel-eslint',
  },
  overrides: [
    {
      files: [
        '**/__tests__/*.{j,t}s?(x)',
        '**/tests/unit/**/*.spec.{j,t}s?(x)',
      ],
      env: {
        jest: true,
      },
    },
  ],
};
```

In this configuration:

- **root: true**
  root: true tells ESLint that this is the root configuration file, meaning ESLint will not look for configuration files higher up in the directory tree.
- **env: { node: true }**
  The env property defines the environments in which the code is expected to run. Setting node: true means that this project is expected to run in a Node.js environment. This enables ESLint to recognize global variables and methods specific to Node.js, such as require, module, etc.
- **globals: { _: 'readonly' }**
  The globals property defines global variables. In this case, the _ variable (which typically refers to the Lodash library) is set as readonly, meaning the value of _ cannot be modified.
- **extends: ['plugin/recommended', 'eslint', '@vue/prettier']**
  The extends property specifies which shared configuration rules ESLint should extend:
  - plugin:vue/recommended: This is a set of recommended ESLint rules for Vue.js projects.
  - eslint:recommended: This is ESLint's built-in recommended rule set, which includes common checks.

    - @vue/prettier: This rule set integrates Prettier with ESLint to ensure that code formatting doesn't conflict with ESLint rules.
- **rules**
  The rules property defines specific ESLint rules:
    - 'no-console': 'off': This rule disables warnings for the use of console. By setting it to 'off', it allows console statements in the code.
    - 'no-debugger': process.env.NODE_ENV === 'production' ? 'error' : 'off': In a production environment (NODE_ENV === 'production'), the use of debugger will result in an error. In non-production environments, it's disabled ('off').
    - 'prettier/prettier': ['error', { singleQuote: true }]: This rule enforces Prettier code formatting with the singleQuote option set to true, meaning it prefers single quotes in the code.
    - 'vue/component-name-in-template-casing': ['error', 'kebab-case']: This rule ensures that Vue component names in templates must be in kebab-case (e.g., my-component). If violated, it will throw an error.
- **parserOptions: { parser: 'babel-eslint' }**
  parserOptions specifies the parser options. In this case, babel-eslint is used as the parser, which allows ESLint to understand the latest ECMAScript features (such as ES6, JSX, etc.).
- **overrides**
  The overrides property allows you to apply different ESLint rules to specific files. Here, the rules are applied specifically to test files:
    - files: ['**/__tests__/*.{j,t}s?(x)', '**/tests/unit/**/*.spec.{j,t}s?(x)']: This specifies the pattern for test files located in __tests__ and tests/unit directories with file extensions .spec.js, .test.js, etc.
    - env: { jest: true }: For these test files, the jest environment is enabled, which makes Jest global variables and functions available.

## 6.3 How to Ignore ESLint Rules

Location: isc-src/web/webui-vue/.eslintignore

The .eslintignore file is a configuration file for ESLint that tells ESLint which files or directories to ignore during the code linting process. This allows developers to skip files that don't need to be checked, such as third-party libraries, generated files, or test files.

**Basic Syntax**

The syntax of .eslintignore is similar to .gitignore and supports wildcards to specify files, directories, or paths to be ignored. Common configurations include:

- **File or Directory Names**: You can directly specify a file or directory name.
- **Wildcards**: Use * or ** to match multiple files or subdirectories.
- **Comments**: You can add comments with # to explain certain rules or directories.
- **Excluding Specific File Types**: You can exclude specific file types by their extensions.

**Example:**

Here are some examples of .eslintignore configurations:

```
# Ignore all .min.js files in the root directory
*.min.js

# Ignore the node_modules directory
node_modules/
```

```
# Ignore the dist directory, usually for compiled files
dist/

# Ignore all .spec.js files in the tests folder
**/tests/**/*.spec.js

# Ignore a specific file
src/legacyFile.js

# Ignore all .html files in the src directory
src/*.html

# Ignore a specific file in the root directory
debug.log
```

**How It Works**

- **Ignoring Directories**: Directories like node_modules/ and dist/ typically contain compiled or external dependencies that don't need linting.
- **Ignoring Specific File Types**: Files like .min.js or .html might not need to be frequently checked or are already minified, so they are typically ignored.
- **Ignoring Test Files**: Test files often have their own style guide, and developers may choose to ignore linting on them, especially if the test framework has its own linting rules.

**Summary**

The .eslintignore file is a simple and effective way for developers to specify which files or directories should not be linted, improving ESLint's efficiency and avoiding unnecessary error reports.